

# Chapter 1

# Categorical Data Analysis

Associate Professor Wikanda Phaphan, Ph.D.





# Introduction

**Categorical data analysis** is the statistical analysis of variables and data whose measurement scale is primarily categorical or discrete.



- **Response Variable:** The primary variable of interest, which is categorical.
- **Observed Values:** These are counts or categorical data, which may be independent or dependent (such as in repeated measures).
- **Distributions:** These may or may not belong to the exponential family. The analysis also involves explanatory variables, which can be either categorical or quantitative.
- **Objective:** To study the structure of relationships among variables and the influence (effects) of explanatory variables on the response variable.



# 1.1 Types of Data



## Data Classified by Properties

- **Quantitative Data:** Data measured as continuous numerical values, such as height, weight, and blood pressure.
- **Categorical Data / Qualitative Data:** Data classified by categories or quality, such as gender, religion, and plant species.



## Data Classified by Collection Method

can be divided into two main types:

- 1. Survey Data:** Information gathered through either a Census (collecting data from every unit in the population) or a Sample Survey (collecting data from a selected group). Its reliability depends heavily on the quality of the sampling design.
- 2. Experimental Data:** Information obtained through controlled testing and direct measurement of outcomes, such as material testing in engineering, clinical trials in medicine, or behavioral observations in social sciences.





## Data Classified by Sources

There are two main types based on where the data originates:

- 1. Primary Data:** Information collected directly from the original source by the researcher for the first time. It is fresh and specific to the task, such as direct results from lab experiments or raw data from a census.
- 2. Secondary Data:** "Second-hand" information obtained from existing records or documents created by others. This includes data from registration records, past research reports, books, or websites.



## Data Classified by Scales of Measurement

1. **Nominal Scale:** Used for classification into groups without a natural ordering (e.g., gender, zip code).
2. **Ordinal Scale:** Used for classification where categories can be ranked or ordered (e.g., pain levels: mild/moderate/severe).
3. **Interval Scale:** Can determine the distance between values but has no absolute zero point (e.g., temperature).
4. **Ratio Scale:** The highest level of measurement; can determine distance and has an absolute zero point (e.g., age, distance).



The four measurement scales can be simplified into two main categories:

- 1. Qualitative Data (Categorical):** This consists of Nominal and Ordinal scales. It is non-continuous and focuses on classification or quality, typically expressed as counts within specific categories.
- 2. Quantitative Data (Numerical):** This consists of Interval and Ratio scales. It is generally continuous and focuses on quantities or magnitudes, representing direct numerical measurements.



## Criteria for Data Collection Design

Once variables are identified, the collection method determines the statistical approach:

- **Survey Data:** Requires Sampling Designs to ensure a representative sample of the population.
- **Experimental Data:** Requires Experimental Designs to control variables and test outcomes.
- **Mixed Data:** Combines both Sampling and Experimental techniques (e.g., sampling units first, then conducting an experiment on them).
- **Pre-existing / Administrative Records:** Analysis depends on the original design used.

Note: If the source or origin is unknown, use the simplest analysis method.





# Methods of Data Collection

Regardless of the data source, researchers must apply specific principles to gather information. Common methods include:

1. **Instruments (Experimental):** Using specialized devices like scales, monitors, or achievement tests; results are documented in a Data Recording Form.
2. **Observation (Surveys):** Counting or tracking specific behaviors (e.g., ATM usage or foot traffic); results are documented in a Data Recording Form.
3. **Interviews (Social Sciences/Business):** Conducting face-to-face, phone, or mail-in sessions using Questionnaires; requires systematically trained interviewers.
4. **Records & Documents (Secondary Data):** Reusing pre-existing information from reports, websites, or registration records; requires a strict Accuracy & Timeliness check.
5. **Expert Judgment (Specialized Analysis):** Relying on the observer's expertise to estimate or categorize complex data (e.g., accents or rainfall); requires High-level Experience.



# From Planning to Analysis

## 1. Planning the Design:

- Sample Surveys Use Sampling Designs (e.g., Multistage Sampling).
- Experiments Use Experimental Designs (e.g., Completely Randomized Design).

## 2. Execution:

- Implement tools (Interviews, Instruments, Observations) to collect data from units.

## 3. Pre-Analysis Check:

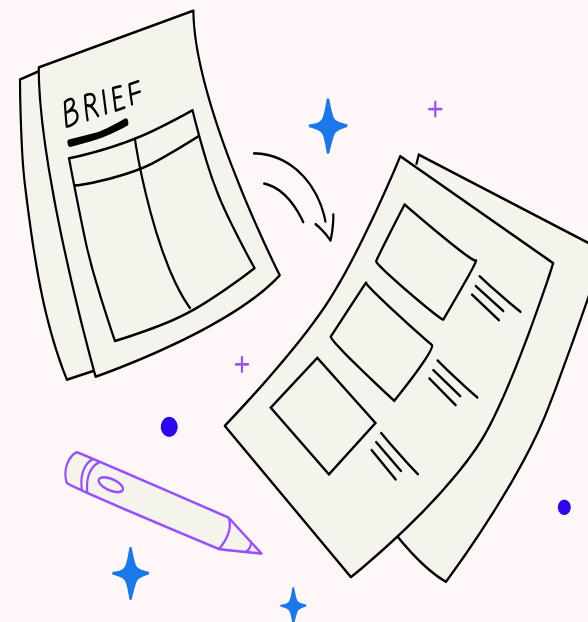
- Before analyzing, examine variable types and data properties to select the most appropriate statistical techniques.



## 1.2 Categorical Variables and Categorical Data

Categorical variables are defined as follows:

**Definition 1.1.** Categorical Variables are non-numeric groups based on qualitative traits. They include Nominal (no order, e.g., gender) and Ordinal (ranked, e.g., severity) types. Additionally, continuous data can be transformed into categories using expert criteria or judgment.





Studio Shodwe

# Types of Variables

1. **Nominal & Ordinal:** Based on qualitative traits or ranks.
2. **Discrete Intervals:** Numbers with very few possible values.
3. **Grouped Continuous:** Continuous data converted into small categories.

**The Discretization Process:** Continuous variables can be categorized through expert judgment or universal criteria to observe frequency (counts).





Measuring categorical data involves counting frequencies within defined qualitative or transformed groups.

**Definition 1.2** Categorical Data The counts (frequencies) of observations within Nominal or Ordinal groups. This includes Interval/Ratio scales that have been transformed into categories based on expert or universal standards.

### Categorical Variables in Modeling:

1. **Response (Dependent) Variable:** Outcomes or results arising from experiments or surveys.
2. **Explanatory (Independent) Variable:** Conditions or groups assigned to the experimental units.



**Table 1.1: Structure of Categorical Data** This table shows **the classification of factor combinations** between **Explanatory variables (Rows)** and **Response variables (Columns)** using counts or frequencies.

| Categories of the explanatory variable | Categories of the response variable |              |     |          | Total    |
|----------------------------------------|-------------------------------------|--------------|-----|----------|----------|
|                                        | 1                                   | 2            |     | c        |          |
| 1                                      | $n_{11}$                            | $n_{12}$ ... | ... | $n_{1c}$ | $n_{1.}$ |
| 2                                      | $n_{21}$                            | $n_{22}$     | ... | $n_{2c}$ | $n_{2.}$ |
| 3                                      | $n_{31}$                            | $n_{32}$     | ... | $n_{3c}$ | $n_{3.}$ |
| ⋮                                      | ⋮                                   | ⋮            | ⋮   | ⋮        | ⋮        |
| r                                      | $n_{r1}$                            | $n_{r2}$     | ... | $n_{rc}$ | $n_{r.}$ |
| Total                                  | $n_{.1}$                            | $n_{.2}$     | ... | $n_{.c}$ | $n_{..}$ |

$n_{ij}$  represent observations or counts, where  $i=1,\dots,r$ , and  $j=1,\dots,c$

$n_{..}$  represent the total number of samples

Studio Shodwe

Examples of categorical variables and categorical data are shown in Table 1.2.

**Table 1.2** Lung cancer classified by daily smoking history

| Smoking history<br>( $X$ ) | Disease status<br>( $Y$ ) |        |       |
|----------------------------|---------------------------|--------|-------|
|                            | Lung cancer               | others | Total |
| none                       | 5                         | 38     | 43    |
| low                        | 8                         | 47     | 55    |
| medium                     | 17                        | 48     | 65    |
| high                       | 7                         | 9      | 16    |
| Total                      | 37                        | 142    | 179   |

Studio Shodwe

Contingency tables are not limited to counts; they can also display probabilities or proportions.

**Table 1.3** Proportion of heart disease patients among examinees classified by blood pressure level and serum

| Cholesterol (CHOL) level<br>(mg/100 cc) | Blood pressure (BP) level |         |         |       |
|-----------------------------------------|---------------------------|---------|---------|-------|
|                                         | 127                       | 127-146 | 146-166 | 167   |
| < 200                                   | 2/119                     | 3/124   | 3/50    | 4/26  |
| 200-219                                 | 3/88                      | 2/100   | 0/43    | 3/23  |
| 219-259                                 | 8/127                     | 11/220  | 6/74    | 6/49  |
| > 259                                   | 7/74                      | 12/111  | 11/57   | 11/44 |





# Response and Explanatory Variables

- **Response Variable (Dependent Variable):** The main variable under investigation, treated as a random variable. In categorical analysis, this variable is always categorical.
- **Explanatory Variable (Independent Variable):** Variables used to explain changes in the response variable. They can be categorical or continuous.

While the response must be categorical, explanatory variables that are continuous can be transformed into categorical ones based on expert criteria to fit specific models like contingency tables.



## categorical variables may have the following types of data



### Nominal Variables

These are qualitative categories without any natural order or hierarchy. Examples include **religion** (Buddhist, Christian, Muslim) and **residence type** (Single house, Townhouse, Condo).



### Ordinal Variables

Categories that follow a natural sequence or ranking, though the exact distance between levels is unknown. Examples include **engine size** (Small, Medium, Large) and **opinion scales** (Strongly disagree to Strongly agree).



### Discrete Interval Variables

Numerical categories with specific, measurable distances but a limited number of values. Examples include **blood pressure levels** or grouped **years of schooling**.



Transforming continuous numerical values into discrete categories for easier observation and frequency counting in contingency tables.

**Table 1.4** Examples of categorical variables of the Discrete Interval type and the Continuous Interval type

| Discrete Interval<br>(Years of schooling:<br>years) | Data<br>(counts) | Continuous Interval<br>(Salary of a company:<br>baht) | Data (counts) |
|-----------------------------------------------------|------------------|-------------------------------------------------------|---------------|
| 1. $\leq 6$ years                                   | 40               | 1. 5,000 – 18,999 baht                                | 35            |
| 2. 7 – 12 years                                     | 50               | 2. 19,000 – 28,999 baht                               | 55            |
| 3. 13 – 15 years                                    | 30               | 3. 29,000 – 38,999 baht                               | 20            |
| 4. $> 15$ years                                     | 20               | 4. 39,000 – 48,999 baht                               | 15            |
|                                                     |                  | 5. 49,000 – 59,999 baht                               | 10            |
|                                                     |                  | 6. $> 60,000$ baht                                    | 5             |



# Statistical Notations & Modeling

- **Primary Symbols:**
  - **Response Variable** (  $Y$  or  $m_{ij}$  ): The main outcome being measured, often representing **Expected Frequencies**.
  - **Explanatory Variable** (  $X$  ): The predictors or factors (can be either qualitative or quantitative).
- **Variable Role in Modeling:**
  - In basic models, we clearly distinguish between **Dependent** (  $Y$  ) and **Independent** (  $X$  ) variables.
  - In **Loglinear Models** (Advanced), all categorical variables are treated as **Factors** to investigate joint effects and associations.



# Example 1.1: Single-Variable Categorical Data

Birthplace of first-year Science students (Academic Year 2009), classified by provinces.

**Table 1.5** Number of new students classified by birthplace.

| Birthplace of new students | Number of new students, Faculty of Science, 2009 |
|----------------------------|--------------------------------------------------|
| Bangkok                    | 120                                              |
| Nakhon Pathom              | 100                                              |
| Ratchaburi                 | 80                                               |
| Suphan Buri                | 90                                               |
| Others                     | 300                                              |
| Total                      | 690                                              |

If an observational unit is classified in **two ways** (e.g., Birthplace + Gender), it becomes **Bivariate Categorical Data**, represented in a **Two-way Contingency Table**



# Example 1.2: Bivariate Categorical Data

Studio Shodwe

Classifying 6 items of sports equipment based on two variables: Color and Shape.

**Table 1.6** Categorical data on the color and shape of sports equipment classified by type of equipment

| Type of equipment | Color variable | Shape variable | Number |
|-------------------|----------------|----------------|--------|
| American football | brown          | oval           | 1      |
| baseball          | white          | round          | 1      |
| golf ball         | white          | round          | 1      |
| basketball        | brown          | round          | 1      |
| badminton         | white          | pointed cone   | 1      |
| hockey            | black          | round cone     | 1      |

If an observational unit is classified in **two ways** (e.g., Birthplace + Gender), it becomes **Bivariate Categorical Data**, represented in a **Two-way Contingency Table**

Studio Shodwe

**Table 1.7** Bivariate categorical data classified by the variables color and shape of sports equipment

| Shape of sports equipment: | Color of sports equipment |       |       |
|----------------------------|---------------------------|-------|-------|
|                            | black                     | brown | white |
| oval                       | 0                         | 1     | 0     |
| round                      | 0                         | 1     | 2     |
| pointed cone               | 0                         | 0     | 1     |
| round cone                 | 1                         | 0     | 0     |

From **Table 1.7**, each category appears only once. The total data for each type of equipment sums to 6 items, which is called a two-way table or two-way contingency table. If the data are classified by more than two variables, it becomes a multidimensional contingency table

The characteristics of categorical variables and examples of categorical variables are summarized as follows:

**Table 1.8** Summary of the characteristics of categorical variables of the discrete type and the continuous type

| Characteristics of variables                      | Examples of categorical variables:                                                                                                                                                         |
|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Categorical with 2 groups<br>(Binary categorical) | male/female state, enterprise/private, fair-skinned/yellow-skinned, military/police, present/absent, recovered/not recovered, successful/unsuccessful, took medicine/did not take medicine |
| Categorical with >2 groups<br>(Nominal scale)     | blood type, cause of death, types of heart disease, color groups, religion, political party, name of university                                                                            |
| Categorical with >2 groups<br>(Ordinal scale)     | severity level of a disease, work performance level, degree level, military–police ranks, academic year level                                                                              |
| Discrete                                          | number of children in a family, number of accidents at an intersection, number of dairy cows that die per year, number of automobiles                                                      |
| Continuous                                        | Height, weight, birth rate, time/unit, income, children’s IQ, annual rainfall                                                                                                              |

Nominal and ordinal variables are generally classified as qualitative data represented by frequencies. In statistical modeling, categorical explanatory variables are referred to as Factors, while their specific subgroups are called Levels (e.g., Gender is a factor with two levels: Male and Female). These variables function as either response or explanatory variables to investigate influences and associations within contingency tables.

**Table 1.9** Two-way contingency table classified by gender and attitude

| Gender | Attitude toward abortion* |    |    | Total |
|--------|---------------------------|----|----|-------|
|        | 1                         | 2  | 3  |       |
| Male   | 11                        | 14 | 6  | 31    |
| Female | 32                        | 28 | 29 | 89    |
| Total  | 43                        | 42 | 35 | 120   |

\*1 represents strongly agree, 2 represents moderately agree, 3 represents disagree (Lawal, 2003)





# 1.3 Population and Sample

**Population in Statistics:** The complete set of all units or measurements within a specified scope of study (not limited to people; can be objects, plants, or data).

- **Types of Population:**

1. **Finite Population:** The total number of units can be clearly identified and counted.
2. **Infinite Population:** The total units are too large or impossible to count (e.g., number of fish in the ocean).

**definition 1.1** A population is the complete set of units under study, while a parameter is a numerical value—such as a proportion ( $P$ ), a mean ( $\mu$ ), a variance ( $\sigma^2$ ) —that directly describes its characteristics.

**Example 1.3** From the U.S. census conducted in 1999 (Watkins et al., 2004), the data of interest concern the number of children per family, with the proportions of families as follows:

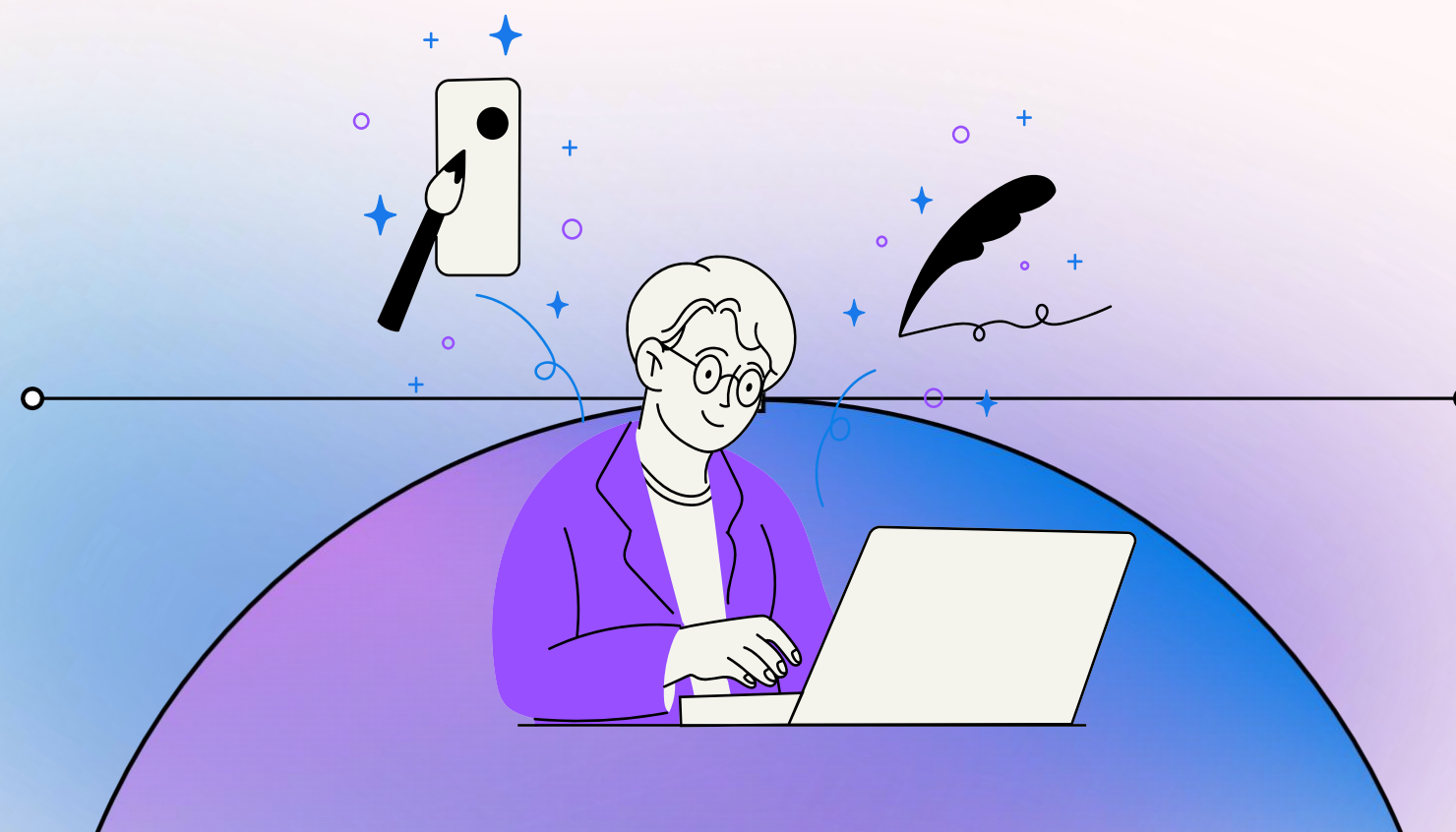
| Number of children per family | Proportion |
|-------------------------------|------------|
| 0                             | 0.505      |
| 1                             | 0.203      |
| 2                             | 0.191      |
| 3                             | 0.073      |
| 4 or more                     | 0.028      |

The values in **Example 1.3** are parameters because they are derived from a national census representing the entire population. Since the data covers every unit in the study area, descriptive statistics are used to summarize the characteristics directly, making inferential methods unnecessary.



Studio Shodwe

A **sample** is a subset of a population used when studying the entire group is impractical. If the sample is a **representative** (randomly selected), **inferential statistics**—such as hypothesis testing and confidence intervals—can be used to draw conclusions about the population. However, if the sample is non-random (e.g., purposive or quota sampling), only **descriptive statistics** should be applied to summarize the sample itself, as it cannot reliably represent the whole population.





**Definition 1.2 A sample** is a subset or a set of data selected from the population of interest. **A statistic** is a formula or a numerical measure that can be computed from the sample data, such as the sample proportion ( $\hat{p}$ ), the sample mean ( $\bar{X}$ ), the sample variance ( $s^2$ ), which are statistics used to estimate the parameter values of the population proportion ( $p$ ), the population mean ( $\mu$ ), and the population variance ( $\sigma^2$ ) respectively.

If **Example 1.3** were based on a representative **sample** instead of a census, the results would be **statistics**, not parameters. In such cases, **inferential statistics** (e.g., hypothesis testing and confidence intervals) must be used to draw conclusions about the population. Ultimately, selecting the correct analytical method depends on identifying both the data type and its source.





# 1.4 Categorical Data Analysis and History

**Evolution of Categorical Data Analysis (GLM)** Historically limited to 2D

Contingency Tables and Chi-square tests, categorical analysis has evolved into Generalized Linear Models (GLMs). GLMs extend beyond normal distributions to handle discrete data within the Exponential Family. A GLM consists of three core components:

1. **Random Component:** The probability distribution of the response variable ( $Y$ ).
2. **Systematic Component:** The linear combination of explanatory variables ( $X$ ).
3. **Link Function:** The function that connects the mean of the response variable to the systematic component.





# Examples of GLMs & Extensions

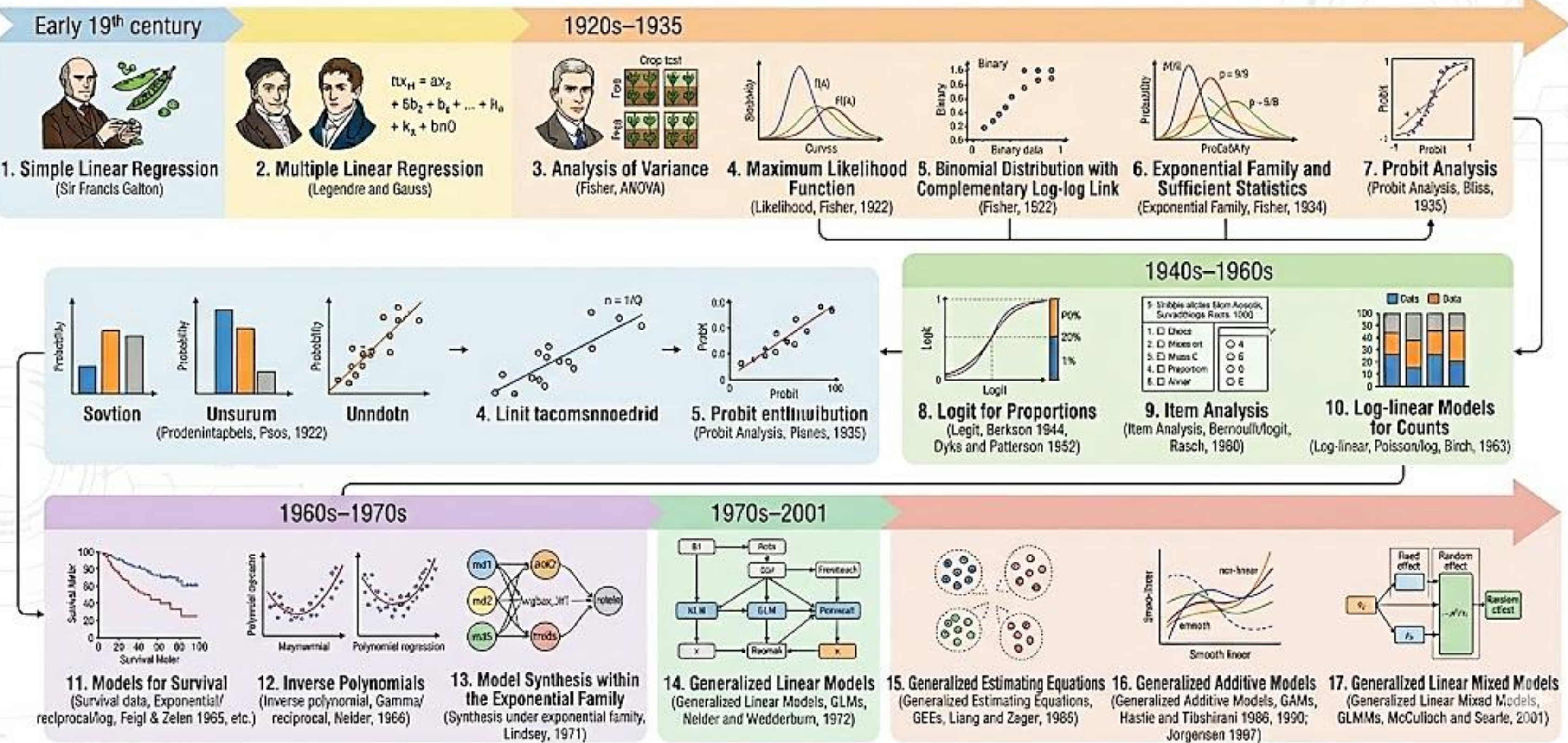
- **Logistic / Probit Models:** Used for binary responses (e.g., Yes/No, Pass/Fail).
- **Multinomial Logit Models:** Used for responses with more than two categories (e.g., Product A, B, or C).
- **Log-linear Models:** Used for analyzing relationships within contingency tables.
- **GLMMs (Generalized Linear Mixed Models):** An extension that incorporates both Fixed Effects (constant across the population) and Random Effects (varying by individual or cluster).





# Simple Linear and Evolution of Linear Models and Statistical Inference

## (THE HISTORY AND EVOLUTION OF LINEAR MODELS AND STATISTICAL INFERENCE)







# Categorical Data Analysis: From Fundamentals to Advanced Modeling

"Modern categorical data analysis has transcended the limitations of traditional statistics by employing diverse statistical modeling techniques, such as **GLMs, GAMs, and GLMMs**. These models are designed to handle non-linear relationships, repeated measurements, and random effects.

The core principle lies in selecting the appropriate **distribution** that fits the data characteristics, such as **Poisson** or **Binomial**. The process typically begins with a preliminary analysis of associations using **Odds Ratios** or contingency table testing. Once a model is constructed, **Maximum Likelihood Estimation (MLE)** is used for parameter estimation. It is then essential to perform a **Goodness of Fit** assessment before interpreting the results to conclude how independent variables influence the response variable. Today, statistical software like **R, SAS, and SPSS** makes these complex calculations both rapid and highly accurate."





# 1.5 Statistical analysis techniques for response variables and explanatory variables

Based on Dobson (1990), the choice of analysis depends entirely on the nature of the **Response** and **Explanatory** variables:

- **Variable Traits:** Categorical data varies by type and measurement scale.
- **Key Decision:** The structural relationship between variables dictates the appropriate statistical model.
- **Reference:** Table 1.9 serves as a standard framework for matching data types with correct analytical techniques.





**Table 1.9** Commonly used statistical analysis techniques classified according to the measurement scales of the response variable and the explanatory variable

| Explanatory variable<br>( $X$ )                       | Response variable ( $Y$ )                                                                        |                                                                                                                                                        |                                                                                                 |
|-------------------------------------------------------|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
|                                                       | 2 categories or levels<br>(Dichotomous or binary,<br>Nominal)                                    | >2 categories or levels<br>(Polychotomous, Nominal –<br>Interval)                                                                                      | Continuous                                                                                      |
| 2 groups                                              | comparison of proportions,<br>contingency table analysis<br>Logistic models, Loglinear<br>models | contingency table analysis<br>Loglinear models, Cumulative logit<br>models, Nonparametric tests                                                        | t-tests<br>correlation coefficient<br>t-tests<br>correlation coefficient<br>Nonparametric tests |
| >2 groups<br>(Nominal)                                | contingency table analysis<br>Generalized logistic models,<br>Loglinear models                   | Loglinear models, Polychotomous<br>logits, Cumulative logit models,<br>Models for matched pairs, GLMs,<br>GAMs, GLMMs, and other mixed<br>models       | ANOVA models<br>Analysis of variance<br>Nonparametric tests                                     |
| Continuous<br><br>Mixed (Categorical &<br>Continuous) | Logistic models<br><br>Logistic models                                                           | Polychotomous logits, Cumulative<br>logit models<br><br>Polychotomous logits, Cumulative<br>logit models, GLMs, GAMs, GLMMs,<br>and other mixed models | Regression analysis<br><br>ANCOVA models<br>Analysis of covariance                              |

GLMs: Generalized Linear Models, GAMs: Generalized Additive Models, GLMMs: Generalized Linear Mixed Models.

**Note:**

- **Variable Counts:** Models can involve multiple response and explanatory variables; techniques should be adjusted accordingly.
- **Scalability:** GLMs can be extended to **Multivariate GLMs** to handle complex structures (e.g., Complex Loglinear Models).



# 1.6 Statistical model building

Categorical Data Analysis extends beyond basic independence tests to **statistical modeling**. By using **Parameter Inference** based on specific distributions (e.g., Binomial, Poisson), we can describe the structural relationships between explanatory and response variables through models like **Logistic Regression, Logit, and Log-linear** models.







# 7 Steps of Statistical Modeling

- 1. Identify Distribution:** Define the response variable's probability distribution.
- 2. Select Variables:** Identify relevant explanatory variables based on domain expertise.
- 3. Build Equations:** Construct plausible models representing the data characteristics.
- 4. Parameter Estimation:** Calculate model parameters.
- 5. Model Selection:** Choose the best-fitting model.
- 6. Model Assessment:** Perform Goodness-of-Fit tests, residual analysis, and hypothesis testing.
- 7. Interpretation:** Evaluate results and summarize findings.

**Note:** Mastery of basic statistics (Hypothesis testing, Regression) and mathematics (Calculus, Matrices) is essential for a deeper understanding of these models.



# 1.7 Probability distributions for categorical data and related sampling schemes

When constructing a statistical model, your first step is to analyze two things:

- **The Response Variable (  $Y$  ):** What is its distributional characteristic? (e.g., Is it a count? A category? A binary choice?)
- **The Explanatory Variables (  $X$  ):** What type of data are they? (e.g., Continuous or Categorical?)





# Exponential Family

In Categorical Data Analysis, we primarily use Generalized Linear Models (GLMs). Most response variables for these models fall into the Exponential Family.

- **Common examples:** Bernoulli (Yes/No), Binomial (Successes in  $n$  trials), Poisson (Counts of events), and Multinomial (Multiple categories).
- **Advanced Models:** While GLMs are the standard, more complex models like GLMMs (Mixed Models) and GAMs (Additive Models) can handle distributions outside this family, though they are harder to interpret.

The way you collect your data (your sampling framework) dictates which probability distribution you must use for your analysis. We can see this clearly in Contingency Tables:

| If your Sampling Scheme is...                                        | The Distribution is... |
|----------------------------------------------------------------------|------------------------|
| Total is random (No fixed sample size)                               | Poisson                |
| Total $i$ is fixed (e.g., exactly 500 people)                        | Multinomial            |
| Row or Column totals are fixed (e.g., exactly 100 Men and 100 Women) | Product Multinomial    |
| Both Row and Column totals are fixed                                 | Hypergeometric         |

Choosing the right method isn't just about clicking buttons in software like SPSS or R. You must understand the nature of your data and how it was sampled. If the sampling design is ignored, your statistical inference (p-values and confidence intervals) may be incorrect.





# Bernoulli Distribution

The Bernoulli distribution is used to describe an experiment or sampling in which the random variable has two possible outcomes: success or failure, independent of one another. Such an experiment is called a Bernoulli trial, with the probability of success equal to  $p$ , and the entire process is called a Bernoulli process.

$$p(y) = p^y q^{(1-y)}; \quad y = 0, 1$$

$$\text{หรือ} \quad p(y) = \begin{cases} p, & y = 1 \\ q, & y = 0 \end{cases}$$

$$\text{โดยที่} \quad 0 \leq p \leq 1, \quad q = 1 - p$$

$$\text{และ} \quad E(Y) = p, \quad \text{var}(Y) = pq$$



# Binomial distribution

The Bernoulli Distribution describes an experiment or sampling where a random variable has only two possible outcomes: Success ( $p$ ) or Failure ( $1 - p$ ). Each observation must be independent of one another.

When multiple observations are conducted under the same conditions, the entire sequence is called a Bernoulli Process. If we consider the sum of the number of successes,  $Y = \sum_{i=1}^n Y_i$ , where the sample size  $n$  is fixed and the probability  $p$  remains constant for all units, the resulting random variable follows a **Binomial Distribution**, denoted as  $Y \sim \text{Binomial}(n, p)$ .



**The probability mass function of the Binomial distribution is**

$$P(Y = y) = \binom{n}{y} p^y (1-p)^{n-y}, \quad y = 0, 1, 2, \dots, n$$

Where  $\binom{n}{y}$  represents the binomial coefficient, which is equal to  $\frac{n!}{y!(n-y)!}$ ,  $y \leq n$

From each Bernoulli trial of  $Y = 0$  or  $1$  obtain  $E(Y_i) = E(Y_i^2) = 1 \times p + 0 \times (1-p) = p$

$$\therefore E(Y_i) = p$$

$$\text{var}(Y_i) = p(1-p)$$

Therefore, the Binomial distribution with  $Y = \sum_{i=1}^n Y_i$  has the following mean and variance:  $\mu = E(Y) = np$  and  $\text{var}(Y) = np(1-p)$  and this distribution converges to normality when  $n$  is large for a fixed



**Notes on Binomial Distribution:** represents a Binomial random variable with parameters  $n$  and  $p$

(1) If  $n = 1$   $Y$  is a Bernoulli random variable with probability of success  $p$

(2) When  $n \rightarrow \infty$  If  $np \geq 5$  and  $np(1-p) \geq 5$ , then  $y$  is approximately distributed as a normal distribution with  $\mu = np$  and  $\sigma^2 = np(1-p)$

(3) When  $n \rightarrow \infty$  If  $p < 0.1$  and  $np < 10$ , then  $y$  is approximately distributed as a Poisson distribution with  $\lambda = np$

(4) Let  $Y_1, \dots, Y_k$  be independent Binomial random variables with parameters  $n_i$  and  $p$ , respectively. The random variable  $Y = Y_1 + \dots + Y_k$  has a Binomial distribution with parameters  $n = n_1 + \dots + n_k$  and  $p$

If sampling **without replacement** from a small population, use the **Hypergeometric Distribution** instead of Binomial, because the observations are **not independent**.





# Multinomial Distribution

A generalization of the Binomial distribution where each trial has  $k$  possible outcomes.

## Notation:

$n$  : Total fixed sample size.

$n_1, n_2, \dots, n_k$  : Observed counts for each category ( $\sum n_i = n$ ).

$\pi_1, \pi_2, \dots, \pi_k$  : Probability of each outcome ( $\sum \pi_i = 1$ ).



The probability mass function of the Multinomial distribution for the counts  $n_j$ ,  $j = 1, 2, \dots, k$  is

$$P(n_1, n_2, \dots, n_k) = \frac{n!}{n_1! n_2! \dots n_k!} p_1^{n_1} p_2^{n_2} \dots p_k^{n_k}$$

Where  $\sum_j n_j = n,$

$$n_k = n - (n_1 + \dots + n_{k-1})$$

$$E(n_j) = np_j,$$

$$\text{Var}(n_j) = np_j(1 - p_j),$$

$$\text{Cov}(n_i, n_k) = -np_j p_k, \quad i \neq j = 1, 2, \dots, k$$



Note:

(1) The marginal distribution of the Multinomial distribution  $n_i$  is Binomial with parameters  $n$  and  $p_i$

(2) If  $k = 2$  and  $p_1 = p$ , **the Multinomial random variable  $n_i$**  becomes a Binomial random variable with parameters  $n$  and  $p$  (the **Binomial distribution** is a special case of the Multinomial distribution when  $k = 2$  )

(3) **The Multinomial distribution can be used in cases where the probabilities  $p_i$ ,  $i = 1, 2, \dots, k$  are equal or unequal**, and it is applied in sampling and advanced statistical analysis.



# Product or Independent multinomial distribution

Analyzed using a **two-way contingency table**, where the **Response variable** (  $Y$  ) is in columns and the **Explanatory variable** (  $X$  ) is in rows.

Assume that  $n_{ij}$  represents the observed count for category  $j$  at level  $i$  of  $X$  (for  $j=1, \dots, k$  and  $i=1, \dots, r$  ). Thus, the counts  $\{n_{ij}, j=1, 2, \dots, k\}$  follow independent Multinomial distributions

**The probability mass function of the Product multinomial is**

$$P(n_{ij}) = \left( \frac{n_i}{\prod_{j=1}^k n_{ij}} \right) \prod_{j=1}^k P_{1|i}^{n_{ij}}$$

and the above  $P(n_{ij})$  is called the “Independent or Product Multinomial Probability Mass Function”.





# Poisson distribution

The Poisson distribution for the counts  $y_i$  that occur randomly within a unit of time, or a specified area or distance (a specific time interval, given region, or space), independently of one another, with a small probability (rare event).

**The probability mass function (by Poisson, 1837) is**

$$P(y) = \frac{e^{-\mu} \mu^y}{y!}, \quad y = 0, 1, 2, \dots$$

Where  $E(Y) = \text{Var}(Y) = \mu$



From Poisson to Normal: When the mean (  $\lambda$  ) increases, the Poisson distribution becomes more symmetrical and begins to resemble the bell-shaped curve of a Normal Distribution.

From Binomial to Poisson: We can use the Poisson distribution to approximate the Binomial distribution only when:

- $n$  (number of trials/people) is very large (e.g., 24 million people).
- $p$  (probability of the event) is very small (e.g., the chance of an accident per person).

Connection Formula: The mean used in the Poisson distribution (  $\lambda$  ) is easily calculated by multiplying the total number of trials by the probability of occurrence (  $\lambda = np$  ).



Note: Let  $Y$  represent a Poisson random variable with parameter  $\lambda$

(1) When  $\lambda \rightarrow \infty$ ,  $Y$  it is approximately distributed as a normal distribution with  $\mu = \lambda$

and  $\sigma^2 = \lambda$

(2) Let  $Y_1, \dots, Y_n$  represent independent random variables with parameters  $\lambda$ , respectively. The random variable  $Y = Y_1 + \dots + Y_n$  follows a Poisson distribution with parameter  $\lambda = \lambda_1 + \dots + \lambda_n$

(3) The waiting time between occurrences of successive Poisson random variable values follows an Exponential distribution (The waiting time between Poisson arrivals is exponentially distributed)



# Hypergeometric distribution

The Hypergeometric distribution is the calculation of 'how many items of interest we are likely to get if we draw a certain number of samples without replacement,' based on the actual total number of items available at the very beginning.

**The probability mass function of the Hypergeometric distribution is**

$$P(y) = \frac{\binom{M}{y} \binom{N-M}{n-y}}{\binom{N}{n}}, \quad y = 0, 1, \dots, n$$

Where  $y \leq M$ ,  $n - y \leq N - M$ ,  $1 \leq n \leq N$ ,  $1 \leq M \leq N$ ,  $N > 0$  is a positive integer and

$$E(Y) = \frac{nM}{N} \quad \text{and} \quad \text{Var}(Y) = \left( \frac{N-n}{N-1} \right) \frac{nM}{N} \left( 1 - \frac{M}{N} \right)$$





Note: Let  $Y$  represent a Hypergeometric random variable with parameters  $n$ ,  $M$  and  $N$

(1) When  $N \rightarrow \infty$  if  $\frac{n}{N} < 0.1$

Then  $Y$  has a Binomial distribution with parameters  $n$  and  $p = \frac{M}{N}$

(2) When  $n$ ,  $M$  and  $N \rightarrow \infty$  if  $\frac{M}{N}$  is small,

Then  $Y$  is approximately distributed as a Poisson distribution with parameter  $\lambda = \frac{nM}{N}$

(3) For example, in a contingency table of size  $(2 \times 2)$  with  $n_1$  and  $n_2$  fixed the distribution of  $n_{ij}$ ,  $j = 1, 2$  depends on  $n_{11}$

**The probability mass function of the Hypergeometric distribution in a contingency table of size  $(2 \times 2)$  is**

$$P(n_{ij}) = \frac{\binom{n_1}{n_{11}} \binom{n_2}{n_1 - n_{11}}}{\binom{n}{n_1}} = \frac{n_1! n_2! n_1! n_2!}{n! n_{11}! n_{12}! n_{21}! n_{22}!}$$



# Negative binomial distribution

From the negative binomial distribution with probability of success equal to  $p$ , if one is interested in the number of failures before observing the  $k$ th success, then the distribution of the number of failures ( $y$ ) is a Negative binomial distribution.

**The probability mass function of the negative binomial distribution is**

$$P(y) = \binom{y+k-1}{k-1} p^k (1-p)^y$$

Where  $y = 0, 1, \dots$ ,  $k = 1, 2, \dots$ ,  $0 \leq p \leq 1$

$$E(Y) = \frac{k(1-p)}{p}, \quad \text{Var}(Y) = \frac{k(1-p)}{p^2}$$



Note: Let  $Y$  have a negative binomial distribution with parameters  $k$  and  $p$

(1) If  $k = 1$  then  $Y$  follows a Geometric distribution with probability of success  $p$

(2) When  $k \rightarrow \infty$  and  $p \rightarrow 1$ , with  $k(1-p)$  fixed, then is approximately Poisson, with mean and variance equal to  $\lambda = k(1-p)$

(3) Let  $y_1, y_2, \dots, y_m$  be independent Negative binomial random variables with parameters  $k_i$  and  $p$  respectively; then the random variable has a negative binomial distribution with parameters  $n = k_1 + k_2 + \dots + k_m$  and  $p$ .



# Exercises for Chapter 1

**1.1** Analyze the following data using an appropriate analytical technique by employing GLIM and SPSS. Assume that the sample data for the random response variable  $Y$  and the explanatory variables  $X_1$ ,  $X_2$ , and  $X_3$  each contain  $n = 10$  observations, and the data are as follows:

|       |      |      |      |      |      |      |      |      |      |      |
|-------|------|------|------|------|------|------|------|------|------|------|
| $Y$   | 13.8 | 16.9 | 19.6 | 23.0 | 25.8 | 28.9 | 30.8 | 33.7 | 35.8 | 42.0 |
| $X_1$ | 23.5 | 23.6 | 23.9 | 24.6 | 25.2 | 25.8 | 26.3 | 26.5 | 26.9 | 26.9 |
| $X_2$ | 23.8 | 23.3 | 24.9 | 28.3 | 31.4 | 34.1 | 37.0 | 38.3 | 42.3 | 45.0 |
| $X_3$ | 16.1 | 16.9 | 17.8 | 18.9 | 20.0 | 21.4 | 22.8 | 24.2 | 25.3 | 27.1 |





**1.2** The model in Question 1.1 can be selected using the Stepwise method with SPSS. That is, after selecting Regression > Linear, enter the dependent variable and the independent variables , then in the same window choose Stepwise instead of Enter, and select OK. The regression model with parameter estimates will be produced by SPSS as required.

**1.3** The model in Question 1.1 can be selected by comparing the differences, or (Change of Deviances), between pairs of models using GLIM, and comparing with If rejected, the shorter model is rejected, and the longer model is accepted accordingly. In addition, the Deviances in Table 1.1 can be used to construct the ANOVA table as shown in Table 1.2.

Table 1.1 Deviances according to models and degrees of freedom (d.f.)

| Model                                               | Deviances | d.f. |
|-----------------------------------------------------|-----------|------|
| $GM(\beta_0)$                                       | 713.8     | 9    |
| $\beta_0 + \beta_1 X_1$                             | 40.45     | 8    |
| $\beta_0 + \beta_1 X_2$                             | 19.77     | 8    |
| $\beta_0 + \beta_1 X_3$                             | 9.078     | 8    |
| $\beta_0 + \beta_1 X_1 + \beta_2 X_2$               | 19.67     | 7    |
| $\beta_0 + \beta_1 X_1 + \beta_2 X_3$               | 9.031     | 7    |
| $\beta_0 + \beta_1 X_2 + \beta_2 X_3$               | 8.949     | 7    |
| $\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3$ | 8.712     | 6    |

Table 1.2 Regression ANOVA

| SOV        | d.f. | SS    | MS    |
|------------|------|-------|-------|
| Regression | 3    | 705.1 | 253.3 |
| Residual   | 6    | 8.7   | 1.45  |
| Total      | 9    | 713.8 |       |



**1.4** From the following data table, construct an ANOVA table using GLIM and SPSS for testing whether there are differences among the 4 types of treatments at the 0.05 level of statistical significance.

| T1   | T2   | T3   | T4   |
|------|------|------|------|
| 54.0 | 51.0 | 53.1 | 50.7 |
| 50.8 | 52.2 | 42.0 | 53.8 |
| 55.0 | 50.0 | 41.5 | 54.9 |
| 47.1 | 53.9 | 50.8 | 55.8 |
| 47.6 | 57.5 | 50.0 | 51.2 |
| 50.0 | 55.2 | 45.8 | 56.7 |

1.5 From measurements obtained to study experimental data for testing 3 types of materials under 4 levels of humidity, suppose the experimental results yield the data shown in the following table:

|                 |   | Humidity (H) |       |       |       |
|-----------------|---|--------------|-------|-------|-------|
|                 |   | 1            | 2     | 3     | 4     |
| Material<br>(M) | 1 | 39.00        | 33.10 | 33.80 | 33.00 |
|                 |   | 42.80        | 37.80 | 30.70 | 32.90 |
|                 | 2 | 36.90        | 27.20 | 29.70 | 28.50 |
|                 |   | 41.00        | 26.80 | 29.10 | 27.90 |
|                 | 3 | 27.40        | 29.20 | 26.70 | 30.90 |
|                 |   | 30.30        | 29.90 | 32.00 | 31.50 |

Analyze the data using GLIM to assist with computation.